Instruction Mode, Note-Taking Method, and Quiz Performance

A 2 x 3 Completely Randomized Factorial Experiment

Presented by:

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1. Research Question

Main research question:

Does instruction mode and note-taking method affect students' quiz performance, and does the effect of note-taking method depend on whether instruction is delivered virtually or in person?

What we wanted to investigate:

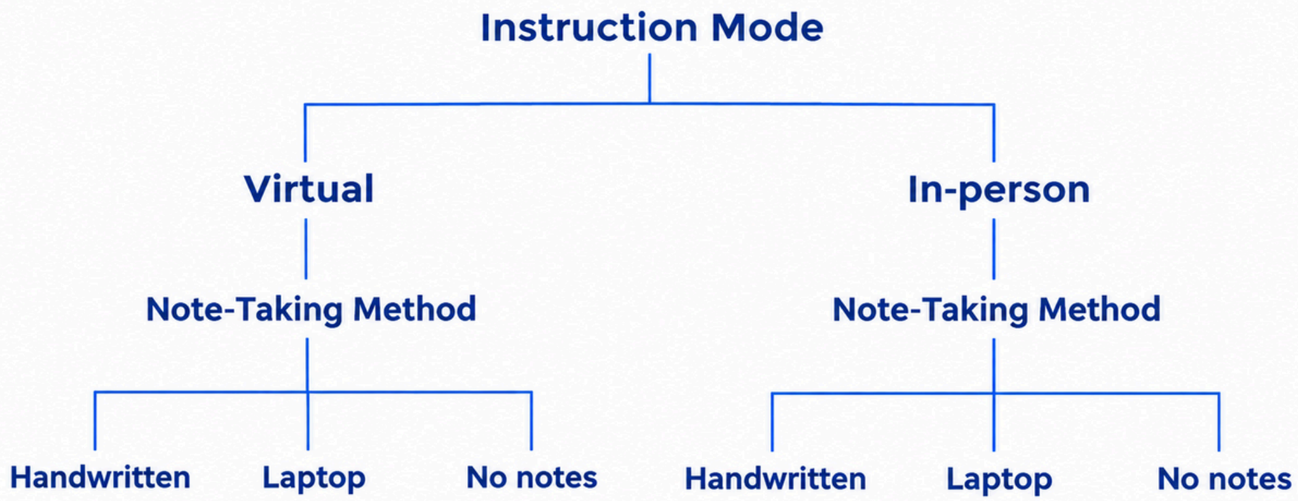
- Whether students perform differently after virtual versus in-person instruction.
- Whether handwritten notes, laptop notes, or taking no notes affect quiz performance.
- Whether the effect of note-taking method changes depending on the instruction mode.



2. Experimental design

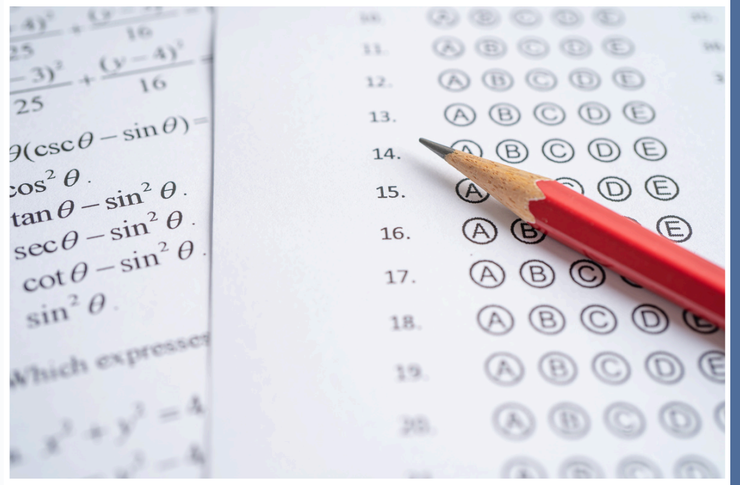
Study Design:

A 2 × 3 completely randomized factorial experiment.



Treatment combinations	6
Participants per combination	12
Response variable	Quiz score from 0 to 10







3. How the experiment was conducted?

- Participants watched an instructional video about recognizing AI-related scams. (https://youtu.be/zb7grp0N7Ek?si=a0gBov-WKrjTdsjY)
- They watched it either virtually or in person, depending on their assigned condition.
- They followed one assigned note-taking method: handwritten notes, laptop notes, or no notes.
- After the video, they completed the same 10-question multiple-choice quiz.
- Each correct answer received 1 point; each incorrect answer received 0 points.
- The total score ranged from 0 to 10.

1. Randomly assign participant
2. Deliver assigned instruction mode
   Virtual or in-person
3. Apply assigned note-taking method
   Handwritten, laptop, or no notes
4. Complete the same 10-question quiz
5. Record quiz score (0–10)

3. How the experiment was conducted?

TEAM FUKURO

VIDEO COMPREHENSION QUIZ

INSTRUCTIONS

Please watch the video carefully and answer all 10 multiple-choice questions based on the information presented.

- You will have _____ minutes to complete the quiz.
- Each question has four possible answers (A, B, C and D).
- Circle or tick the letter corresponding to your chosen answer.
- Select only ONE answer for each question.
- Please do your best when answering all questions.

QUESTIONS

1. According to the video, why are AI scams becoming harder to detect?

A. They are only used by large companies.
B. AI-generated content is becoming more realistic.
C. AI can no longer generate images.
D. People rarely use the internet.

2. If you receive a phone call from a friend asking for emergency money, what should you do first?

A. Transfer the money immediately.
B. Ignore the call.
C. Verify that your friend is actually in trouble before sending money.
D. Post about it on social media.

3. What is a deepfake?

A. A secure online password.
B. AI-generated audio, video, or images designed to imitate a real person.
C. A type of computer virus.
D. A private messaging app.

4. According to the video, which of the following was once a common clue that an image was AI-generated?

A. Perfect spelling.
B. Too many fingers or oddly shaped hands.
C. Bright colors.
D. High image quality.

3. How the experiment was conducted?

student_id	instruction_mode	note_taking	quiz_score
S001	Virtual	Handwritten	7
S002	Virtual	Handwritten	8
S003	Virtual	Handwritten	7
S004	Virtual	Handwritten	6
S005	Virtual	Handwritten	6
S006	Virtual	Handwritten	9
S007	Virtual	Handwritten	5
S008	Virtual	Handwritten	8
S009	Virtual	Handwritten	8
S010	Virtual	Handwritten	10
S011	Virtual	Handwritten	7
S012	Virtual	Handwritten	8
S013	Virtual	Laptop	8
S014	Virtual	Laptop	8
S015	Virtual	Laptop	2
S016	Virtual	Laptop	6
S017	Virtual	Laptop	5
S018	Virtual	Laptop	8
S019	Virtual	Laptop	9

4. Hypotheses

We tested three effects using a significance level of $\alpha=0.05$.

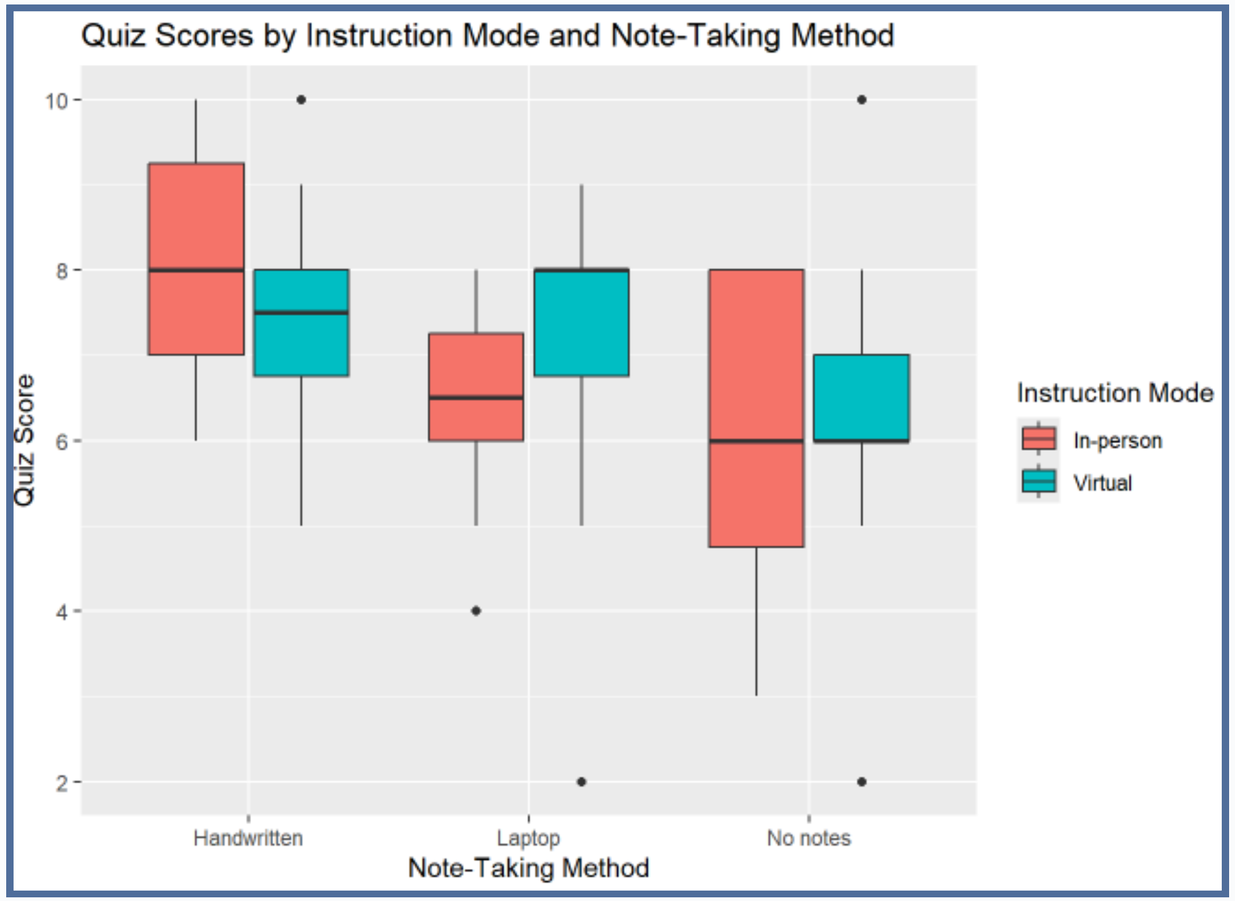
Effect being tested	Null hypothesis (H0)	Alternative hypothesis (Ha)
Instruction mode	Mean scores are equal for virtual and in-person instruction.	Mean scores differ by instruction mode.
Note-taking method	Mean scores are equal for all three note-taking methods.	At least one note-taking mean differs.
Interaction	The effect of note-taking method does not depend on instruction mode.	The effect of note-taking method depends on instruction mode.



5. Descriptive Results

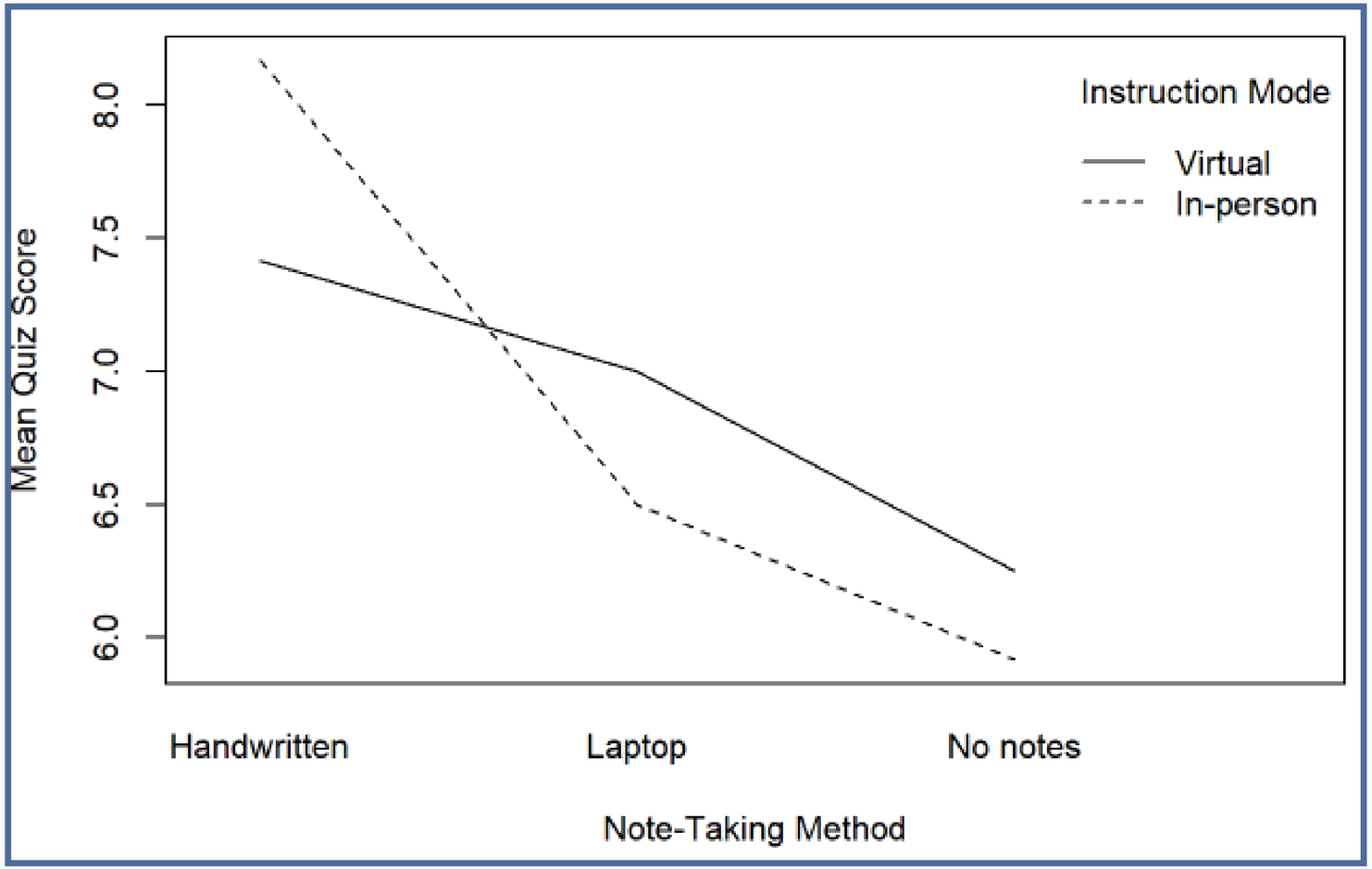
Instruction mode	Note-taking method	Participants	Mean score	SD
Virtual	Handwritten	12	7.42	1.4
Virtual	Laptop	12	7	1.9
Virtual	No notes	12	6.25	1.9
In-person	Handwritten	12	8.17	1.5
In-person	Laptop	12	6.5	1.2
In-person	No notes	12	5.92	1.8

6. Distribution of Quiz Scores



Comparison of the spread and median of quiz scores across the six treatment groups.

7. Interaction graph



The lines suggest some interaction, but the interaction test was not significant. Therefore, the effect of note-taking method did not significantly differ by instruction mode.

8. Two-way ANOVA results

A significance level of 0.05 is used for all three tests.

```
##                                Df Sum Sq Mean Sq F value  Pr(>F)
## instruction_mode                1   0.01   0.014   0.005 0.94248
## note_taking                     2  35.58  17.792   6.720 0.00221 **
## instruction_mode:note_taking    2   5.53   2.764   1.044 0.35783
## Residuals                      66 174.75   2.648
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

The significant F-test for note-taking method indicates that a difference exists among the three methods, but it does not identify which methods differ. Tukey's procedure is used to identify the significant pairwise differences.

9. Post-hoc analysis

Mean Score

```
## # A tibble: 3 × 4
##   note_taking     n  mean    sd
##   <fct>       <int> <dbl> <dbl>
## 1 Handwritten    24  7.79  1.44
## 2 Laptop         24  6.75  1.59
## 3 No notes       24  6.08  1.79
```

As the note-taking main effect was significant, Tukey's procedure was used to determine which note-taking methods differed.

Tukey's HSD Test

```
##   Tukey multiple comparisons of means
##     95% family-wise confidence level
##
## Fit: aov(formula = quiz_score ~ instruction_mode * note_taking, data = quiz_data)
##
## $note_taking
##                               diff       lwr          upr     p adj
## Laptop-Handwritten    -1.0416667 -2.167934  0.08460026 0.0755950
## No notes-Handwritten -1.7083333 -2.834600 -0.58206641 0.0015527
## No notes-Laptop       -0.6666667 -1.792934  0.45960026 0.3369481
```

Handwritten notes had the highest mean quiz score (7.79), followed by laptop notes (6.75) and no notes (6.08). Tukey's test found that only handwritten notes versus no notes was statistically significant ($p = 0.0016$); instruction mode and the interaction effect were not significant.

10. Conclusion



01

Note-taking method mattered. The ANOVA found statistically significant differences among the three note-taking methods.

02

Handwritten notes performed best descriptively. The average score was 7.79, compared with 6.75 for laptop notes and 6.08 for no notes.

03

The clearest pairwise difference was handwritten notes versus no notes. Tukey's test found this difference statistically significant.

04

Instruction mode was not statistically significant. We did not find sufficient evidence of a difference between virtual and in-person instruction.

05

The interaction was not statistically significant. We did not find sufficient evidence that the effect of note-taking method differed by instruction mode.

11. Limitations and Improvements

Limitations

- Prior knowledge of AI-related scams was not measured.
- The dataset did not include a separate measure of whether participants followed their assigned note-taking condition.
- Virtual and in-person environments may have differed in distractions, equipment, sound quality, or supervision.
- There were 12 participants per treatment combination, so smaller effects—especially interactions—may have been difficult to detect.

Possible Improvements

- Give participants a short pretest to measure prior knowledge.
- Record whether participants followed their assigned note-taking condition.
- Standardize video delivery, instructions, equipment, timing, and quiz conditions more carefully.
- Conduct a pilot study to test the procedure and quiz difficulty.
- Recruit more participants per treatment combination.

Thank You!